2.3.10 Phylum: Hemichordata

(G.*hemi* –half; *chorda*-string)

Hemichordates were earlier treated as a subphylum of Chordata (or Prochordata). They are now regarded to be an independent phylum of invertebrates, close to Echinodermata. The animals of this group possess the characters of invertebrates as well as chordates.

This phylum consists of a small group of worm-like, soft marine animals, mostly tubiculous and commonly called the **'acorn worms'** or **'tongue worms'**. They are bilaterally symmetrical, triploblastic and coelomate animals with organ system level of organisation. Their body is cylindrical and is divided into three regions, the anterior proboscis, a short collar and a long trunk. Most hemichordates are ciliary feeders. Their circulatory system is simple and open or lacune type with a dorsal heart. Respiration is through paired gill slits opening into the pharynx. Excretion is by a single **proboscis gland** or **glomerulus** situated in the proboscis. Nervous system is primitive. Sexes are separate and exhibit sexual mode of reproduction; Fertilization is external. Development is indirect with a free swimming **tornaria larva**.

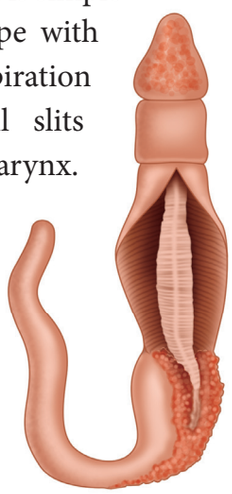


Figure 2.18 Example of Hemichordata - Balanoglossus

Examples: *Balanoglossus*, *Saccoglossus, Ptychodera flava (*Indian Hemichordate found in Kurusadai islands in Tamilnadu) (Figure 2.18).

2.4 Phylum: Chordata

(G. *Chorda –string*)

Chordata is the largest phylum with most familiar group of animals, such as fishes, amphibians, reptiles, birds and mammals and less known forms such as **lancelets** (Amphioxus) and **tunicates** (Ascidian). All chordates possess three fundamental distinct features at some stage of their life cycle (Figure 2.19), they are:

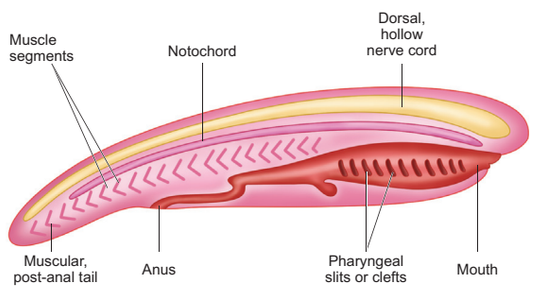


Figure 2.19 A Typical Chordate

1. Presence of elongated rod like notochord below the nerve cord and above the alimentary canal. It serves as a primitive internal skeleton. It may persist throughout life in lancelets and lampreys. In adult vertebrates, it may be partially or completely replaced by backbone or vertebral column.
2. A dorsal hollow or tubular fluid filled nerve cord lies above the notochord and below the dorsal body wall. It serves to integrate and co-ordinate the body functions. In higher chordates, the anterior end of the nerve cord gets enlarged to form the brain and the posterior part becomes the spinal cord, protected inside the vertebral column.
3. Presence of pharyngeal gill slits or clefts in all chordates at some stage of their lifecycle. It is a series of gill slits or clefts that perforates the walls of pharynx and appears during the development of every

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